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Unit-III
Corrosion Science and Management

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Outline

Sl No	Topic to be taught	Duration
1	Introduction to corrosion, Electrochemical theory of corrosion	1 hour
2	Types of corrosion: differential aeration (pitting and water line), differential metal and stress corrosion.	1 hour
3	Factor affecting rate of corrosion-electrode potential (nature of the metal), Nature of corrosion product, numerical problems on PB ratio.	1 hour
4	Factor affecting rate of corrosion -continue-pH of the medium, temperature, relative anodic and cathodic area (Area effect).	1 hour
5	Corrosion control: Metal coating-galvanization and tinning, surface conversion coating - anodizing and phosphating.	1 hour
6	Cathodic protection - sacrificial anode method. Corrosion testing by weight loss method. Corrosion penetration rate (CPR)-numerical problems	1 hour
7	Metal finishing: Introduction to Electroplating, electroplating of chromium.	1 hour
8	Introduction to electroless plating, electroless plating of copper.	1 hour
9	Case studies on corrosion failure.	Self learning

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Outline

- Metals and alloys have become an integral part of any engineering- steel for civil engineer, aluminum and its alloys for automobile engineer, copper for an electrical or electronics engineer and so on.

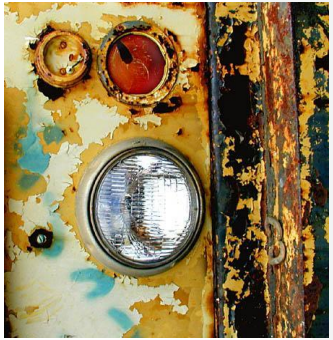
- All this has become possible because of their properties via. Conductivity, malleability and ductility and characteristic luster.

- Nature has given us lot more of Iron, aluminum, copper etc., which tend to corrode. As they have a natural tendency to corrode, they must be protected.

- Then comes the question-how to protect them?

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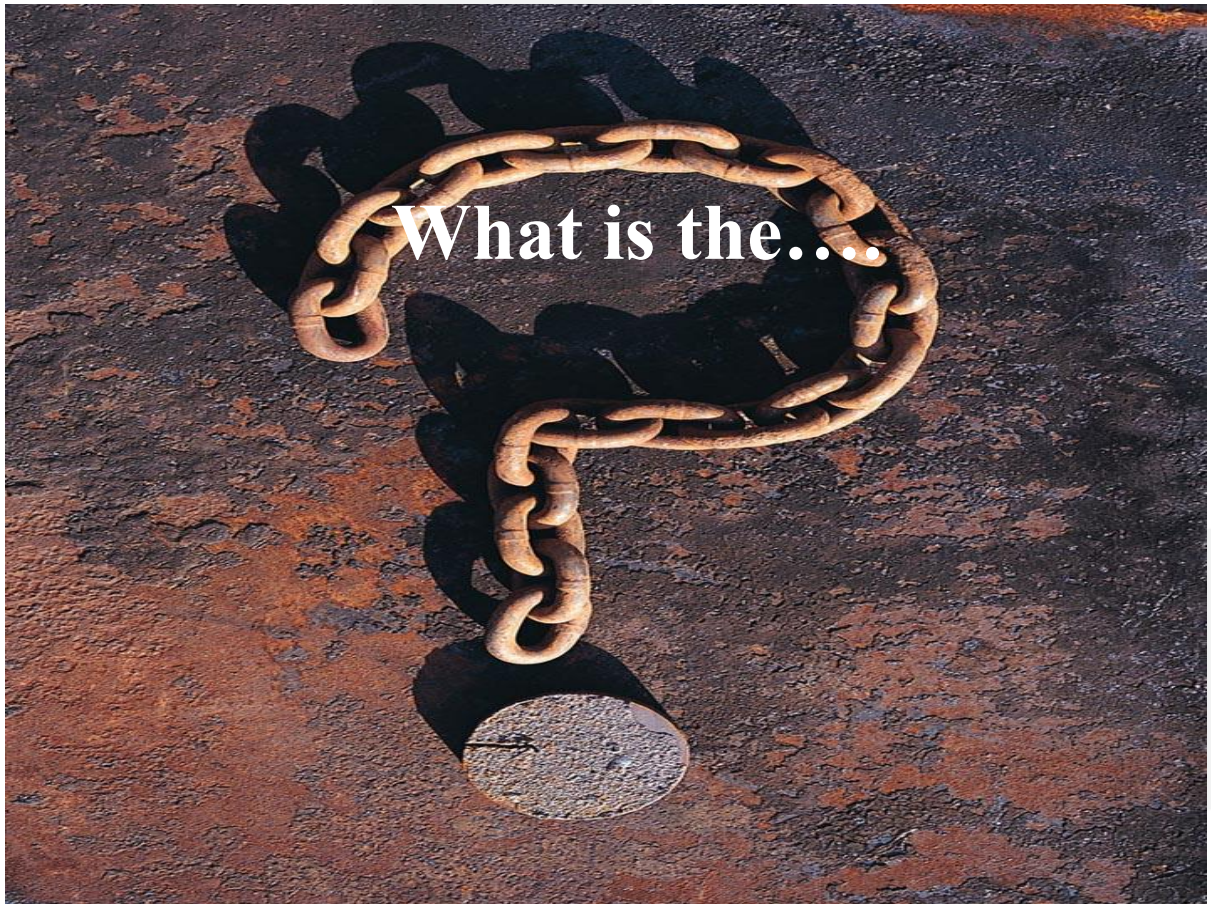




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Cost of Corrosion?



Corrosion causes a Global Annual Damage of 2.9 Trillion USD !!!!!

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Significance of Corrosion on Infrastructure



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Effect of corrosion on civil structures



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Engineer finds corrosion in collapsed bridge at North Carolina speedway (2000)



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Why Metals Corrode?

The Corrosion product is much stable than the pure metal.

For example:

Iron oxide (Hematite) is more stable than Iron,

Copper sulfate is more stable than copper

Corrosion can be defined as the deterioration of material by reaction to its environment. Corrosion occurs because of the natural tendency for most metals to return to their natural state; e.g., iron in the presence of moist air will revert to its natural state, iron oxide.

Types of corrosion:

1)**Dry corrosion:** Corrosion in the absence of moisture/aqueous medium.

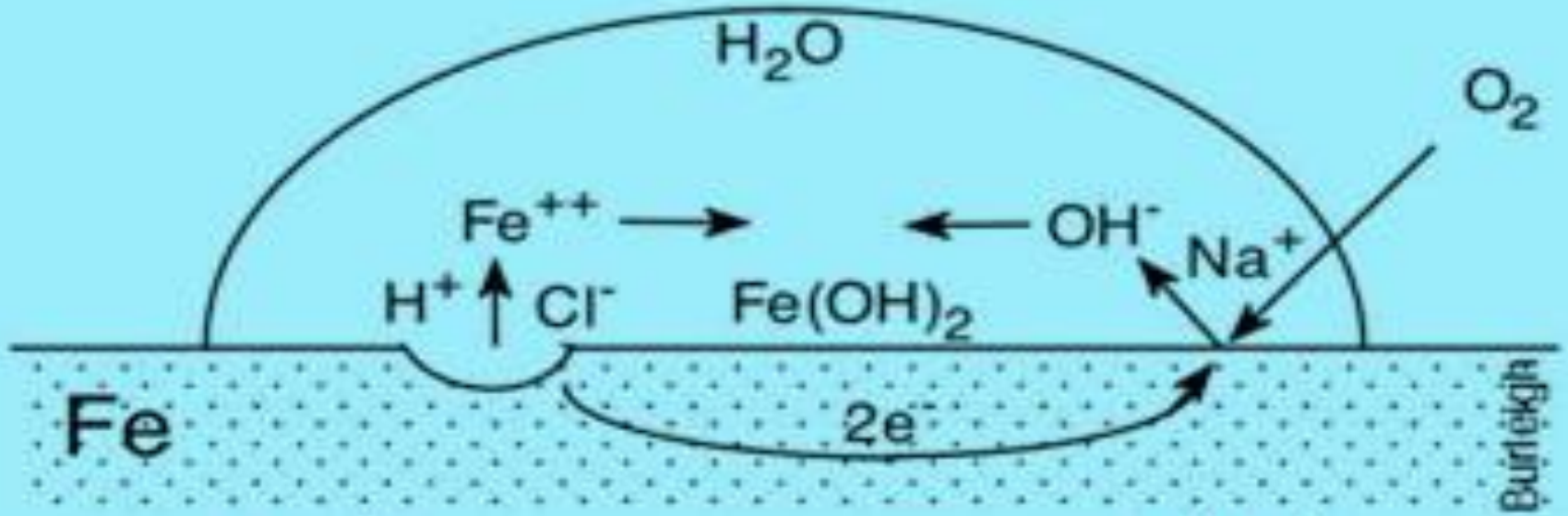
It is a direct chemical attack of atmosphere gases like CO_2, SO_2, Cl_2, etc. or to strong acids (H_2SO_4), HCl , Alkali. on Metals,

Oxidation by Oxygen in absence of moisture

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Wet Corrosion

- Corrosion in presence of moisture (Electrochemical theory of corrosion)



$Fe \rightarrow Fe^{++} + 2e^-$

$2H_2O + O_2 + 4e^- \rightarrow 4OH^-$

$Fe^{++} + 2H_2O \rightarrow Fe(OH)_2 + 2H^+$

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Electrochemical Theory

- When a metal is exposed to moist aerated environment, tiny galvanic cells are formed.
- Portion of metal behaves as anode (impurity, stress, deformation, difference in concentration of air …)
- 3 required components in an electrochemical corrosion cell: 1) An anode; 2) A cathode; 3) A conducting environment for ionic movement (electrolyte).
- If any of the above components is missing or disabled, the electrochemical corrosion process will be stopped.
- Anode undergoes oxidation Eg: $Fe \rightarrow Fe^{+2} + 2e^{-}$
- Cathode directly doesn't involve in the reaction, but Cathodic reactions are taking place in the surrounding.

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Electrochemical Theory

Anodic reaction

$$Fe \longrightarrow Fe^{2+} + e^-$$

Cathodic reactions

	H_2 Evolution (absence of O_2)	O_2 Absorption (Presence of O_2)
Acidic (pH<7)	$2H^+ + 2e \longrightarrow H_2$	$2H^+ + 1/2\ O_2 + 2e \longrightarrow H_2O$
Basic/Neutral (pH≥7)	$2H_2O + 2e \longrightarrow 2OH^- + H_2\uparrow$	$2H_2O + 1/2\ O_2 + 2e \longrightarrow 2OH^-$

$Fe \longrightarrow Fe^{2+} + 2e^-$ (At anodic area)

$2H_2O + O_2 + 4e^- \longrightarrow 4OH^-$ (At cathodic area)

$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2$ (insoluble and Unstable)

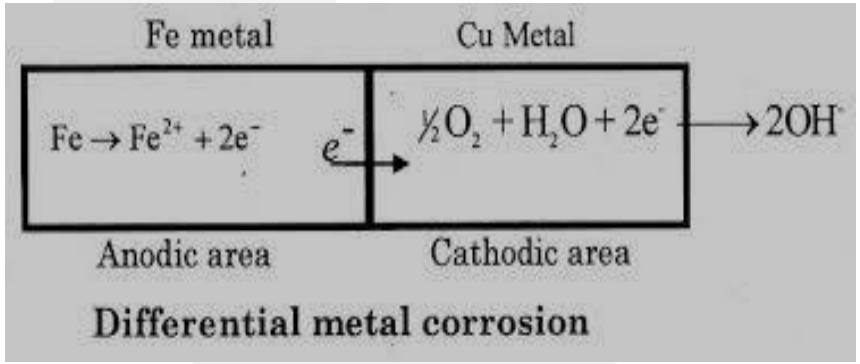
$Fe(OH)_2$ is not stable . It gets further oxidized by oxygen

$4Fe(OH)_2 + O_2 + 2H_2O \longrightarrow 4\ Fe(OH)_3$ or $Fe_2O_3.3H_2$ (Rust)

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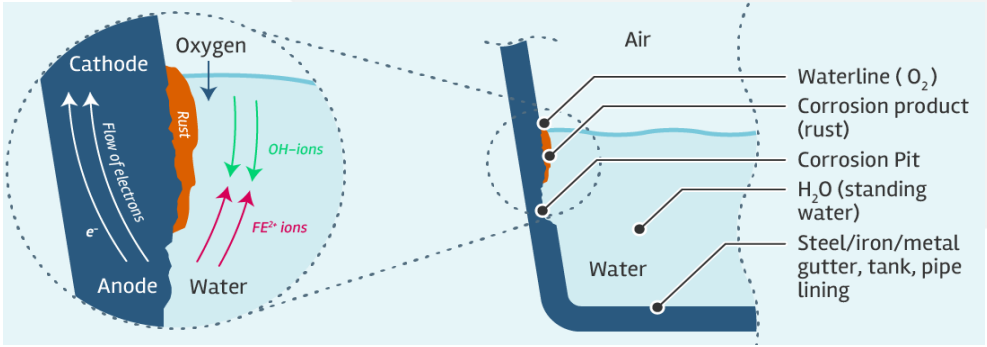
Types of Corrosion

1. Differential metal Corrosion (Galvanic Corrosion)



Differential metal corrosion

2. Differential Aeration Corrosion



Differential Aeration Corrosion

- This type of corrosion occurs when one part of the metal is exposed to a different air concentration from the other part.
- This causes a difference in potential between differently aerated areas.
- It is found that **less oxygenated part acts as anode more oxygenated part acts as cathode.**

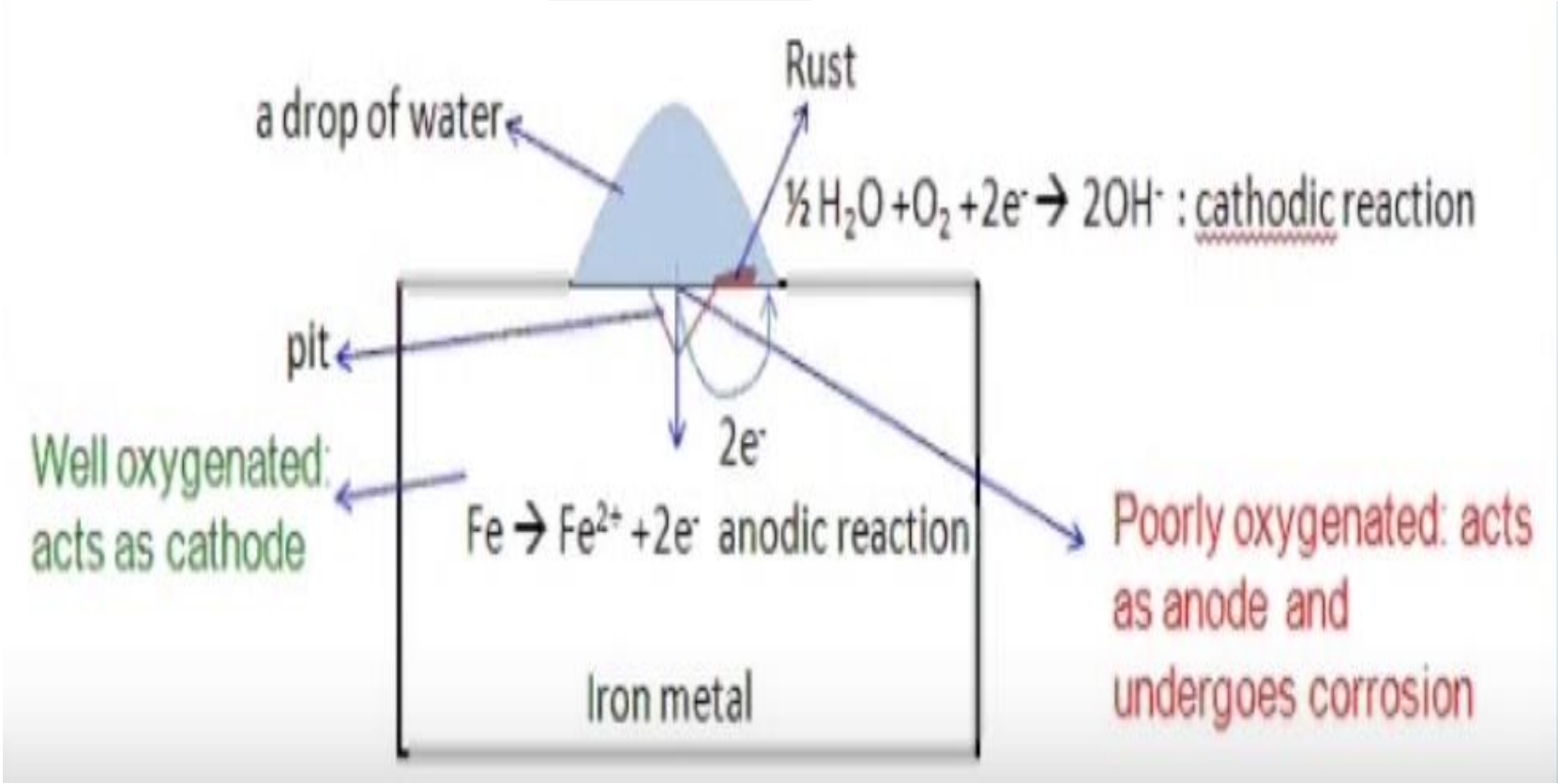
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Differential aeration corrosion





Differential Aeration Corrosion



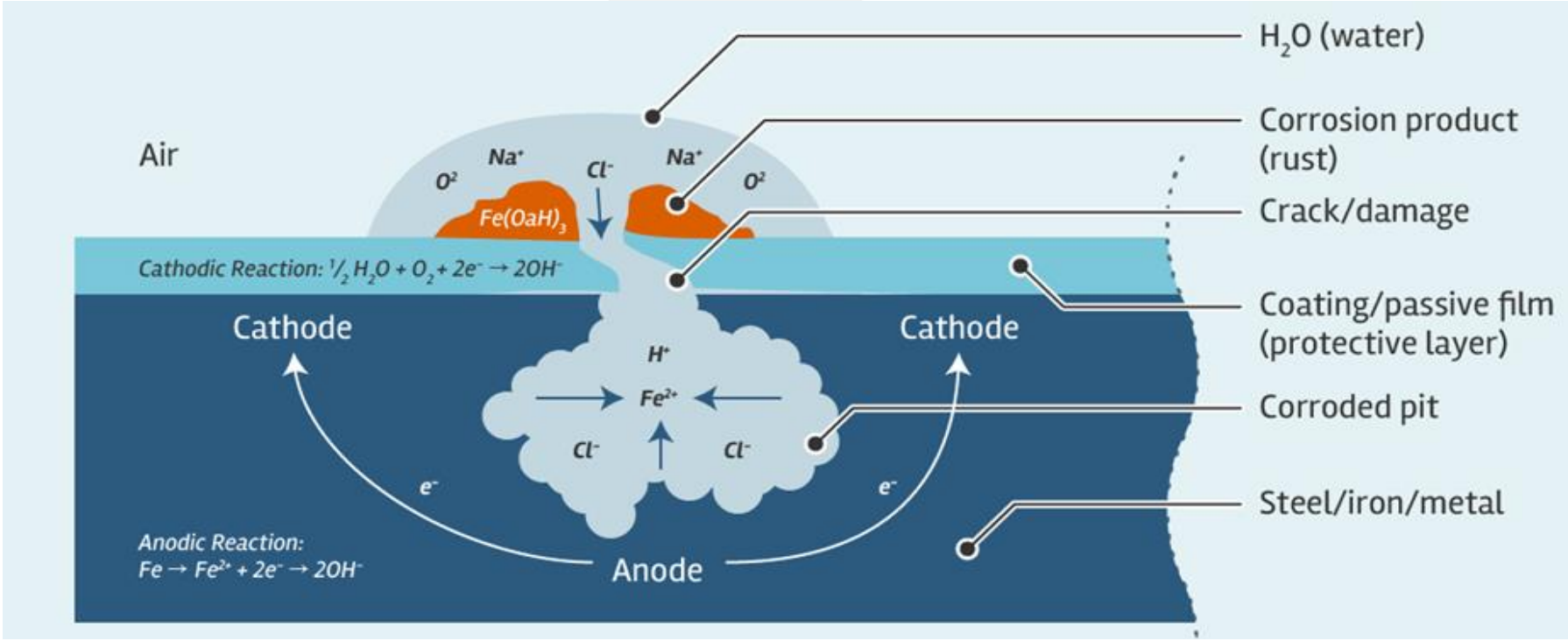
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Differential Aeration Corrosion

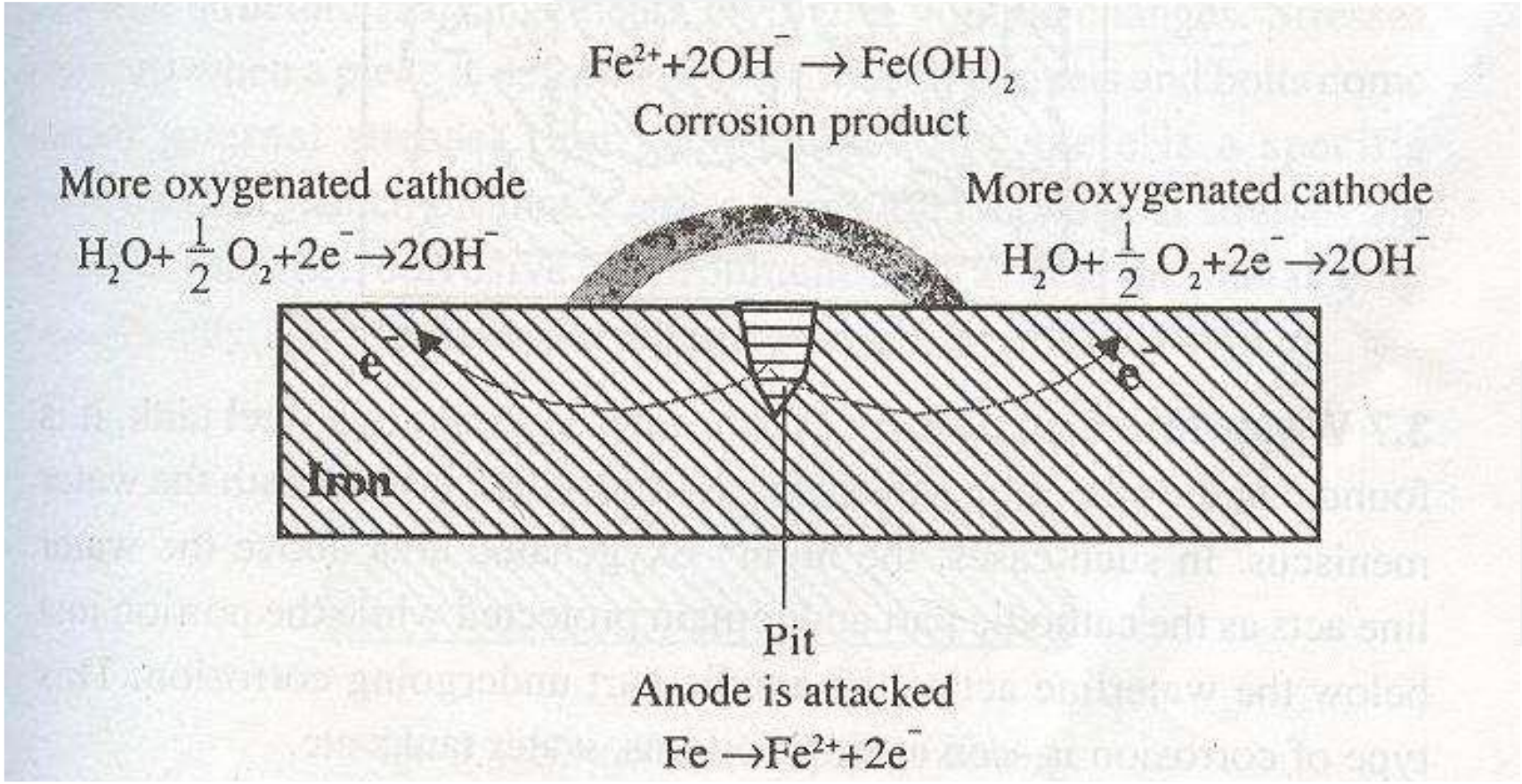


Pitting Corrosion

- The localized and intense corrosion of a metal leading to the formation of a pit or pin hole around which the metal is relatively unattacked, is called as pitting corrosion



Pitting Corrosion



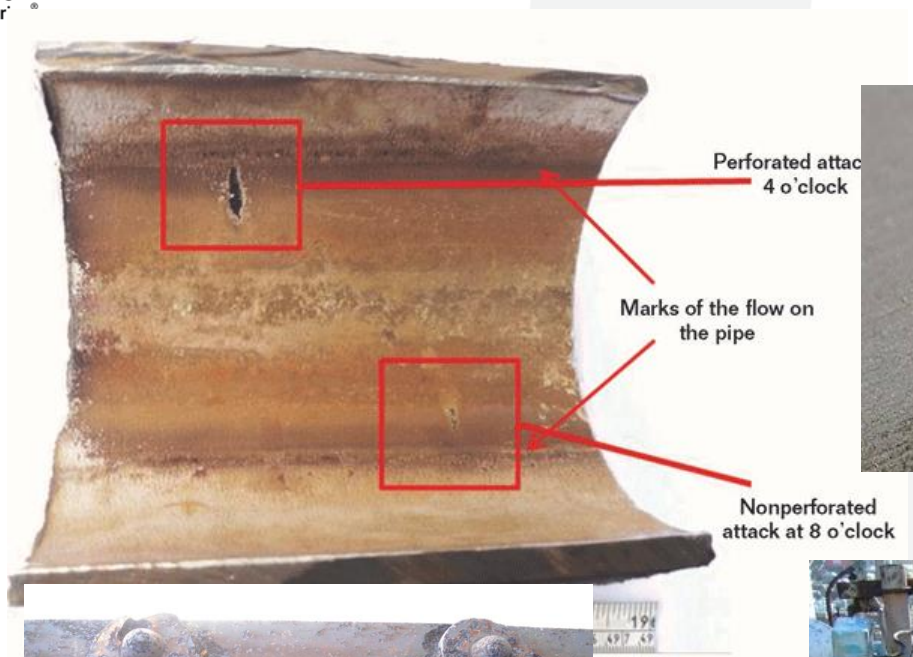
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Pitting Corrosion



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Pitting Corrosion

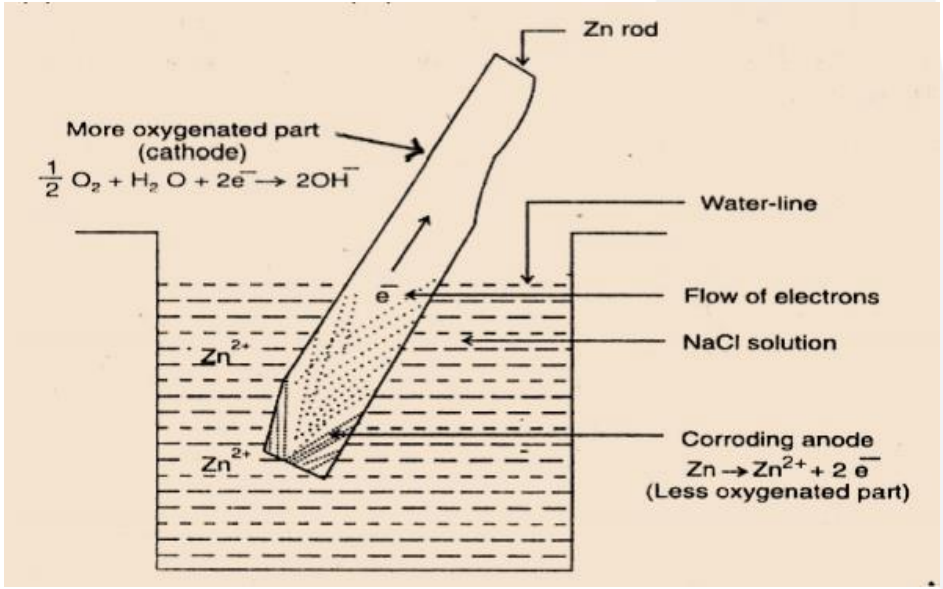


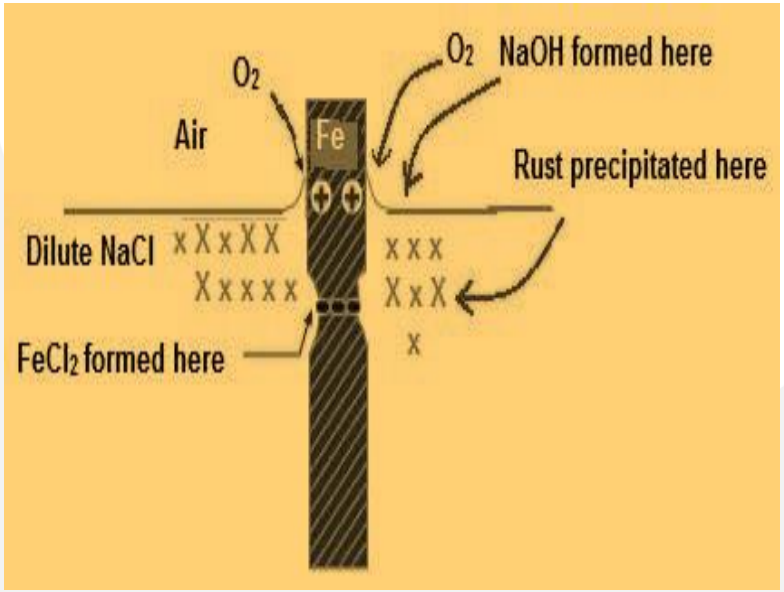






Waterline Corrosion





The part of the metal just below the waterline (less aerated) acts as anode and corrodes. The part above the waterline acts as cathode. Corrosion product is usually observed below the water level.

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Effect of Waterline Corrosion

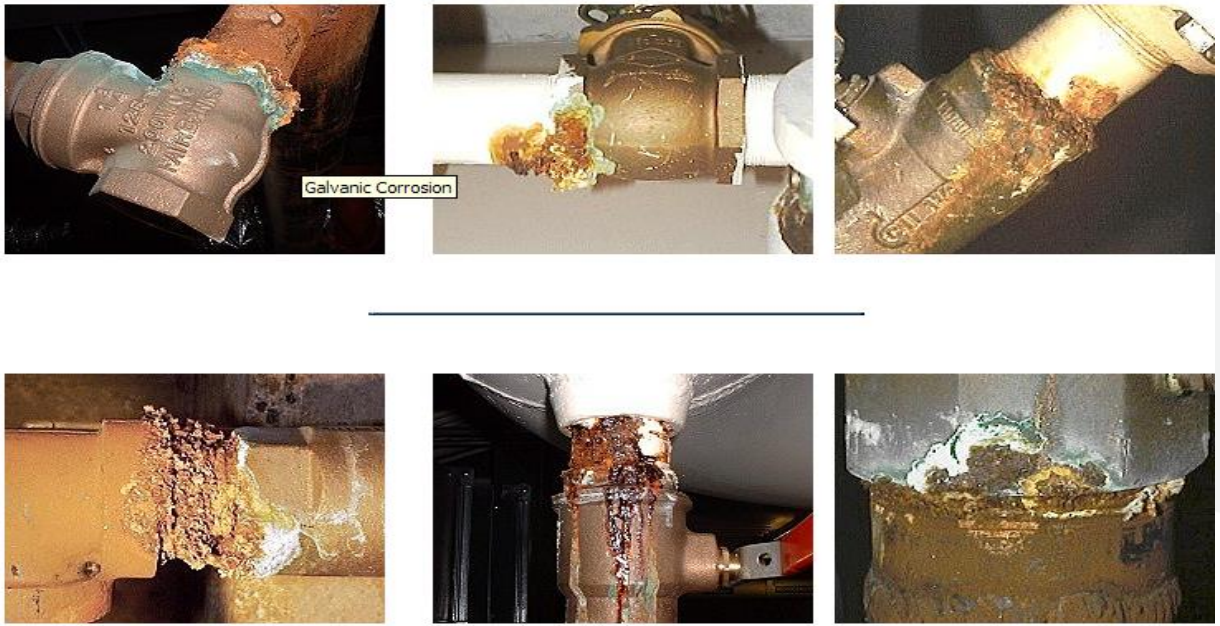






Galvanic Corrosion

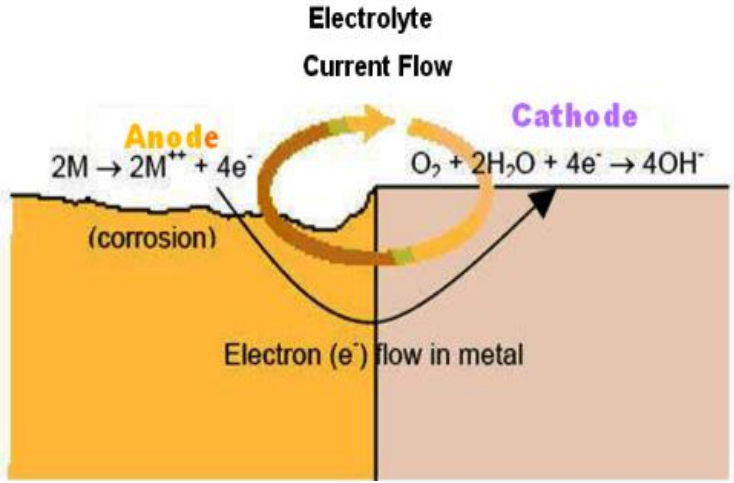
- Dissimilar metals are in contact and in the presence of an electrolyte.
- The more anodic metal corrodes.



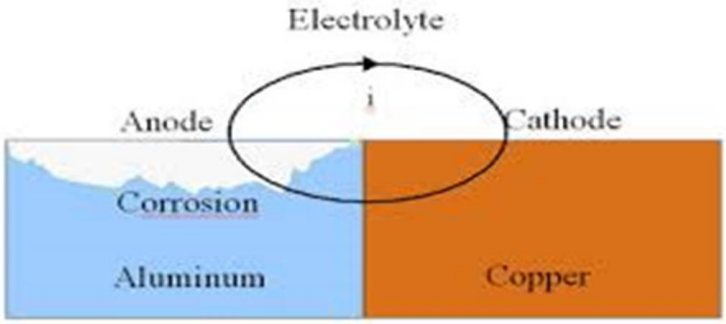


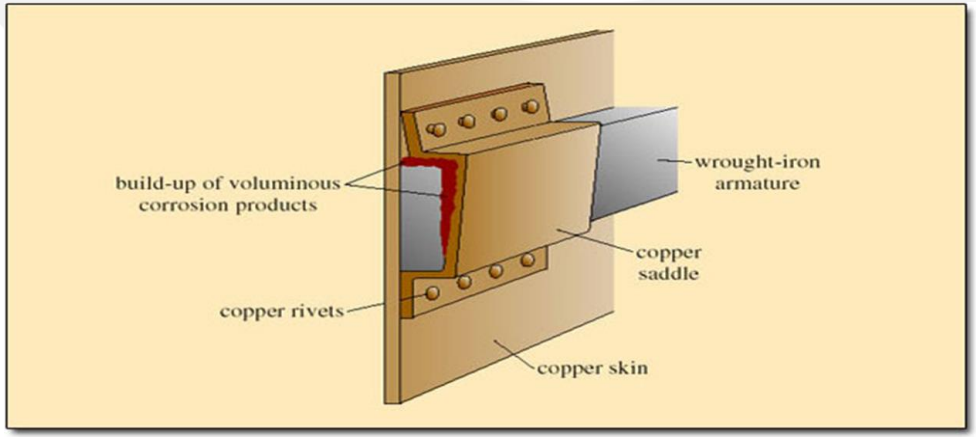
Bilge pump - Magnesium shell cast around a steel core.

Galvanic Corrosion

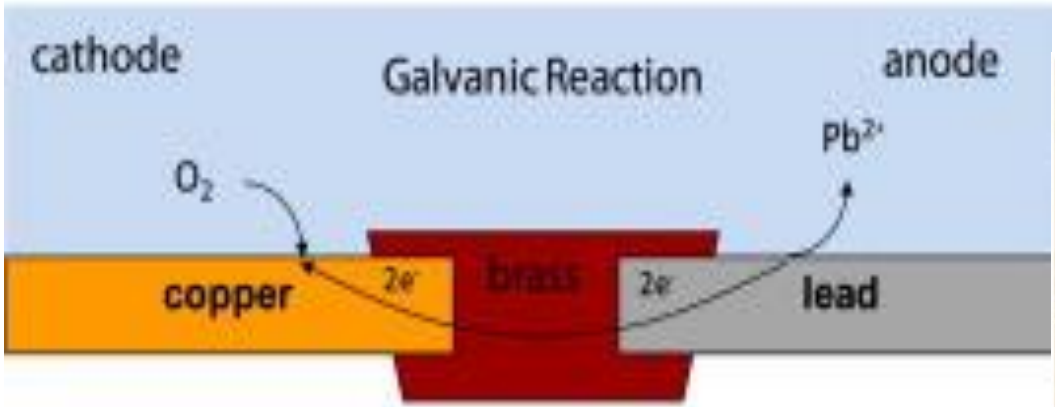


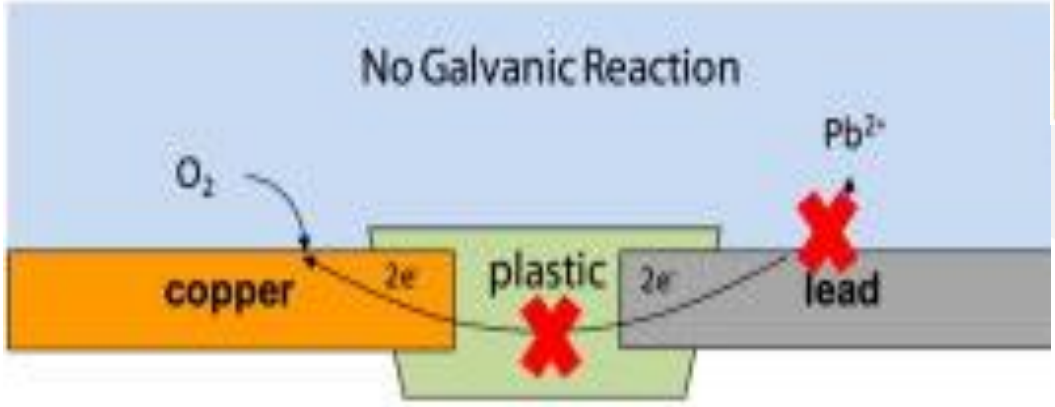


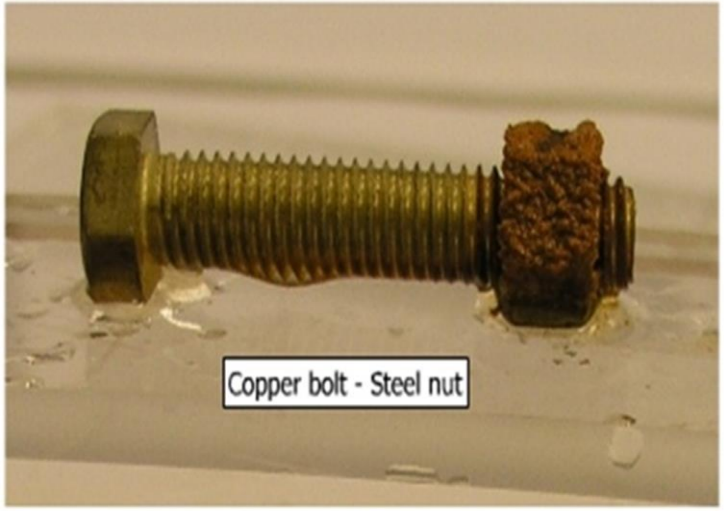




Galvanic Corrosion







Galvanic Corrosion

During Galvanic corrosion,

- One of the metals having lower reduction potential becomes anode.
- The other metal acts as cathode. The reaction at cathode depends on nature of the environment.

The rate of corrosion depends on,

- Difference in electrode potentials. Larger the difference, more is the rate of corrosion.
- Relative areas of anode and cathode.
- Tendency of a metal to undergo passivation.
- Environmental factors like pH, temp, conc of oxygen.

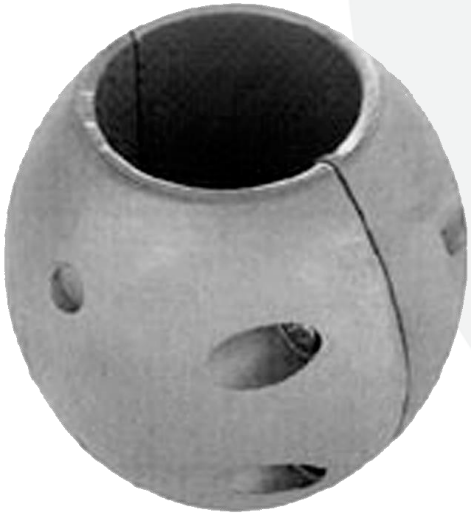
Galvanic Corrosion

Galvanic corrosion can be minimized by,

- Selecting metals that are very close in galvanic series, while designing the metallic structure.
- Selecting metals such that cathodic area is small and anodic area is large.
- Preventing the contact between two metals through proper insulation using insulating washers.

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Results of Galvanic Corrosion



New Zinc
(for 1” diameter shaft)



Old Zinc after 8 months
(for 1” diameter shaft)

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Results of Galvanic Corrosion





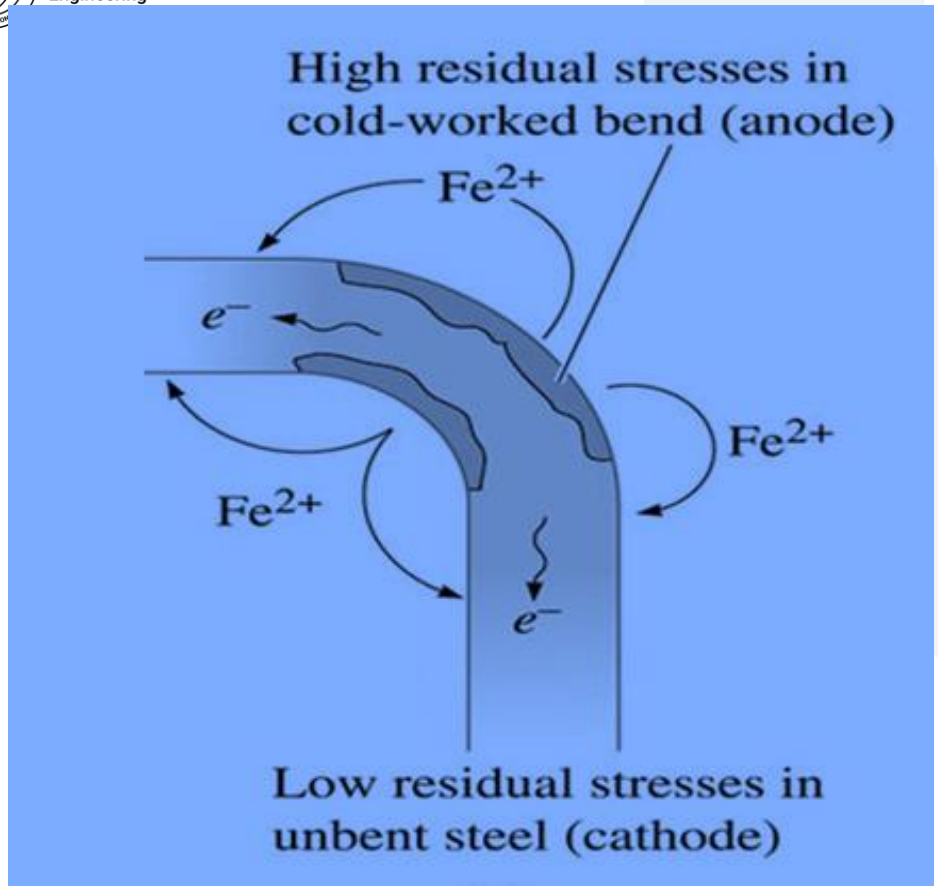




Stress Corrosion

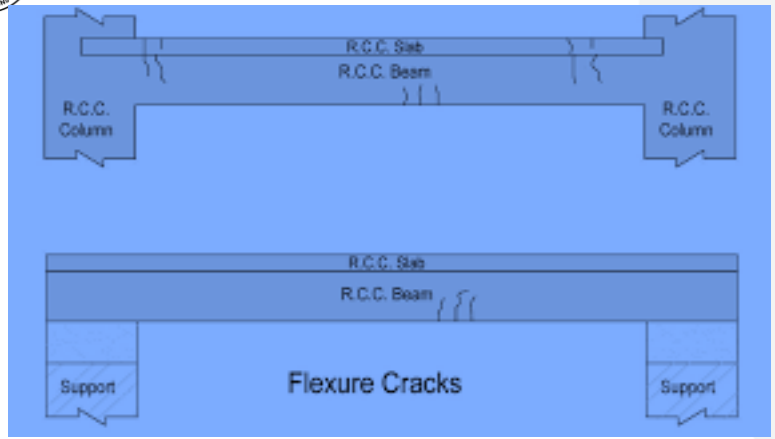
- When the metal/alloy under ***uneven mechanical load or stress***, that leads to formation of Anodic and cathodic area.
- The area under stress is anodic area, and the area not under stress is cathodic area.
- Metal atoms under stress are compressed in their crystal system, to nullify the stress they reduce their size by undergoing oxidation.

Stress Corrosion





Stress Corrosion









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Factors affecting Corrosion

1. **Nature of metal**

a) Physical state of metal
b) Purity of metal
c) Difference in SRP Value
d) Surface finishing
e) Defect in the metal

2. Nature of corrosion product

Passivation,

Solubility of corrosion product

3. Area effect

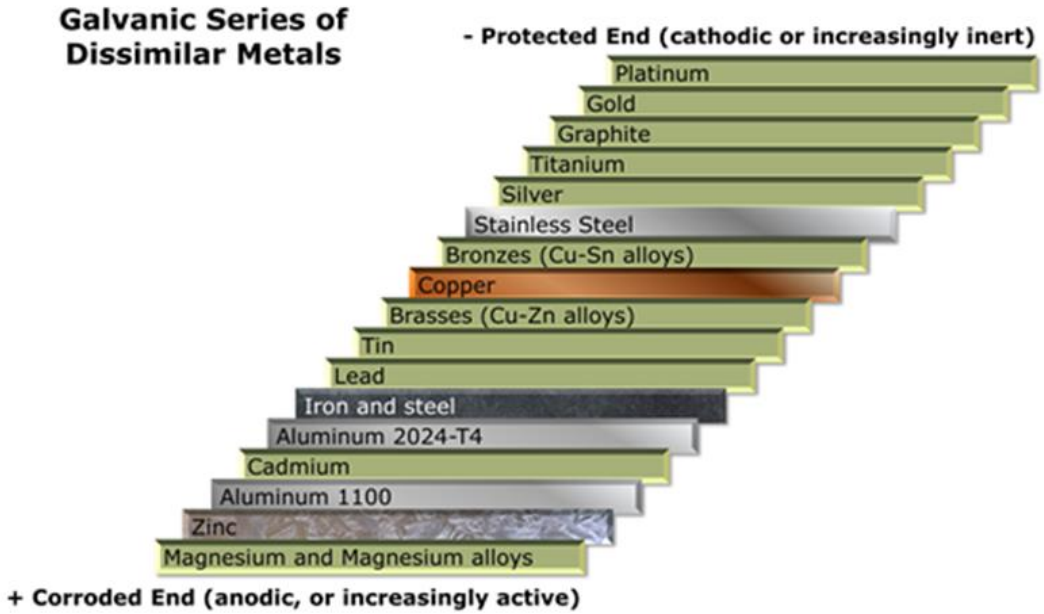
Relative anodic and cathodic area

4. Nature of Environment

a) Temperature
b) Moisture
c) pH
d) Gasses

Nature of Metals

- Metals like Na, K, Mg, Zn etc. corrode faster than Cu, Ni, Ag etc in air. Lower the reduction potential, higher will be its tendency to get oxidised (corroded).
- SRP of Fe is -0.44V whereas that of Cu is +0.34V. Fe corrodes faster than Cu.
- This will not hold good for metals like Al, Ti, Cr etc that gets passivated.



Nature of Corrosion Product

- Corrosion occurs only when a suitable environment exists.
- When there is a barrier between the metal and its environment, that prevents the contact between the two, corrosion stops.
- If such barrier is created by the metal itself, then it is said to undergo passivation.

If the corrosion product is	
Soluble	Insoluble
Volatile	Non-volatile
Non-adherent	Adherent
Non-stoichiometric	Stoichiometric
Porous	Non-porous
Conducting	Insulator
More CORROSION	***No/Less CORROSION***
Ex: Fe_2O_3	***Ex: Al_2O_3***

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Nature of Corrosion Product

Specific volume ratio= Vol of metal oxide / Vol of Metal

Pilling Bedworth rule

If Volume of metal oxide $\geq$ volume of metal

Oxide layer is protective or non-porous

e.g. *Al*

If Volume of metal oxide $<$ volume of metal

Oxide layer is porous

e.g. *Alkali and alkaline earth metals*

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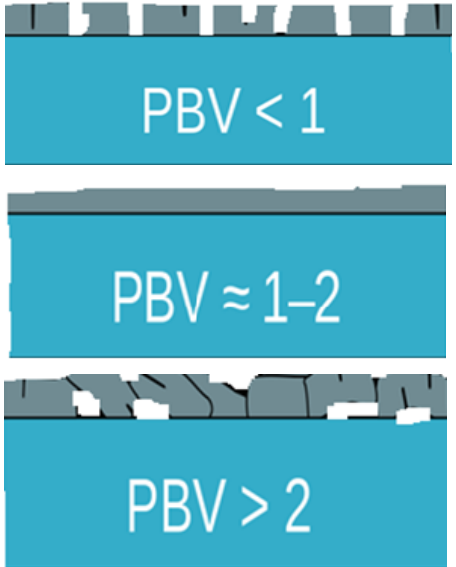
Passivation

Pilling Bedworth rule:

- The Pilling–Bedworth ratio (P–B ratio) is the ratio of the volume of the metal oxide to the volume of the corresponding metal (from which the oxide is created).

$$R_{PB} = \frac{V_{oxide}}{n \cdot V_{metal}} = \frac{M_{oxide} \cdot \rho_{metal}}{n \cdot M_{metal} \cdot \rho_{oxide}}$$

M- Atomic or molecular mass
n- No of metal atoms per mole of metal oxide
ρ - Density



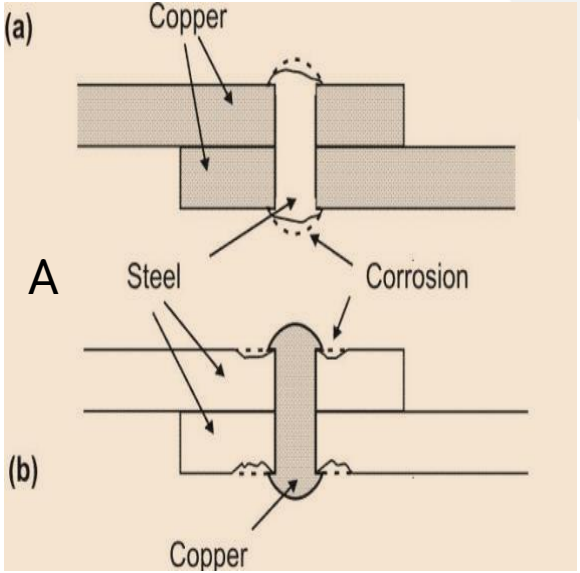
1. What is the nature of oxide film on Mg. Given, density of MgO=3.65 mg/m^3, Mg=1.74 mg/m^3. The mol wt of MgO=40.3 and atomic wt of Mg=24.3

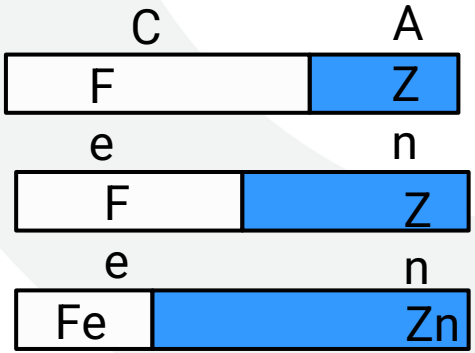
   Ans: 0.79 porous (unprotective).

Area Effect

- If there is larger cathodic area and a small anodic area on a surface, the rapid oxidation occurs on small anodic area leading to rapid increase in the corrosion.

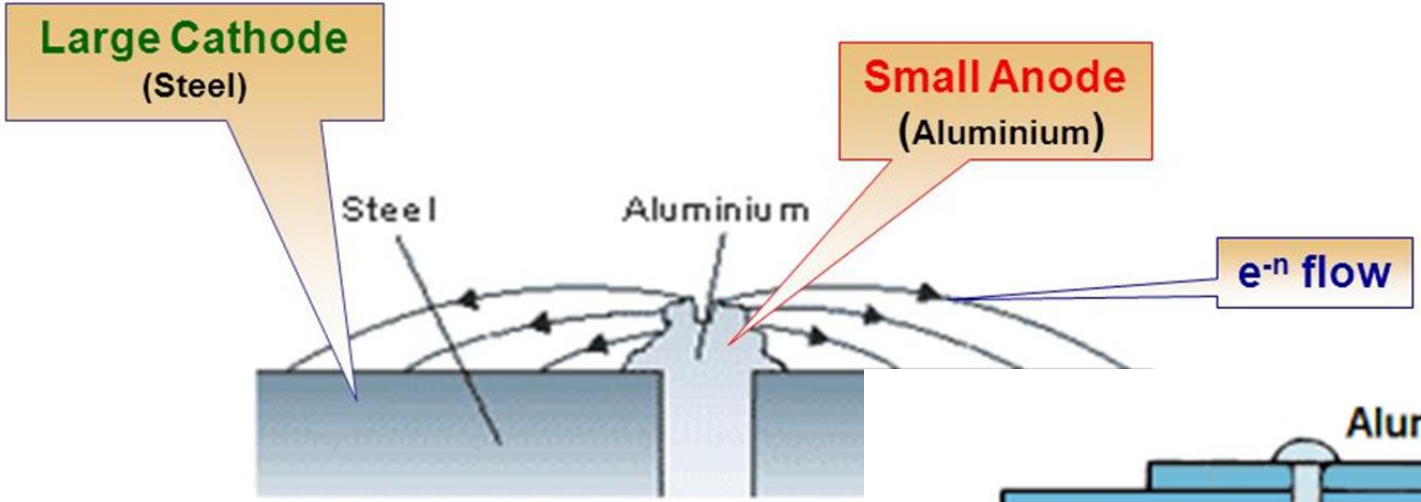
- If cathodic area is small and anodic area is large, slow and uniform corrosion occurs.



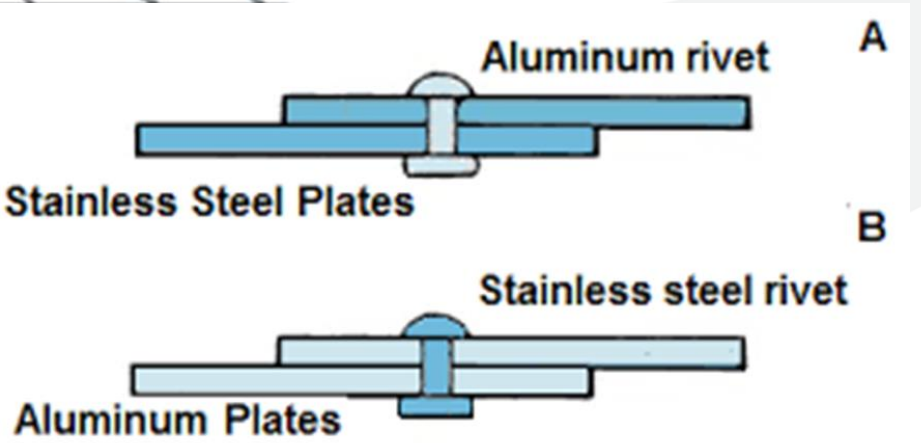


Area Effect

Rate of corrosion is more if the size of cathodic area is large



If cathodic area is larger, the demand f
this results in the increase of rate of oxi
high rate of corrosion at anode.



Nature of the Environment

i) Temperature: If the temperature increases,

- **conductivity of the medium increases**
- **adherence property of the film decreases**
- **passivation decreases**
- **volatility of the corrosion product increases**
- **solubility of the corrosion product increases**

Hence corrosion also increases.

Nature of the Environment

(iii)Presence of impurities in atmosphere: due to CO_2, H_2S, SO_2 etc in the vicinity of industrial area.

(iv)Effect of pH:

Lower the pH, higher the acidic, higher the concentration of H^+ ion. Higher the rate of corrosion

$$2H^+ + 2e \longrightarrow H_2$$

Exception: Al/Al_2O_3 (Amphoteric),

if pH<7, Al_2O_3 film is very stable, no further corrosion.

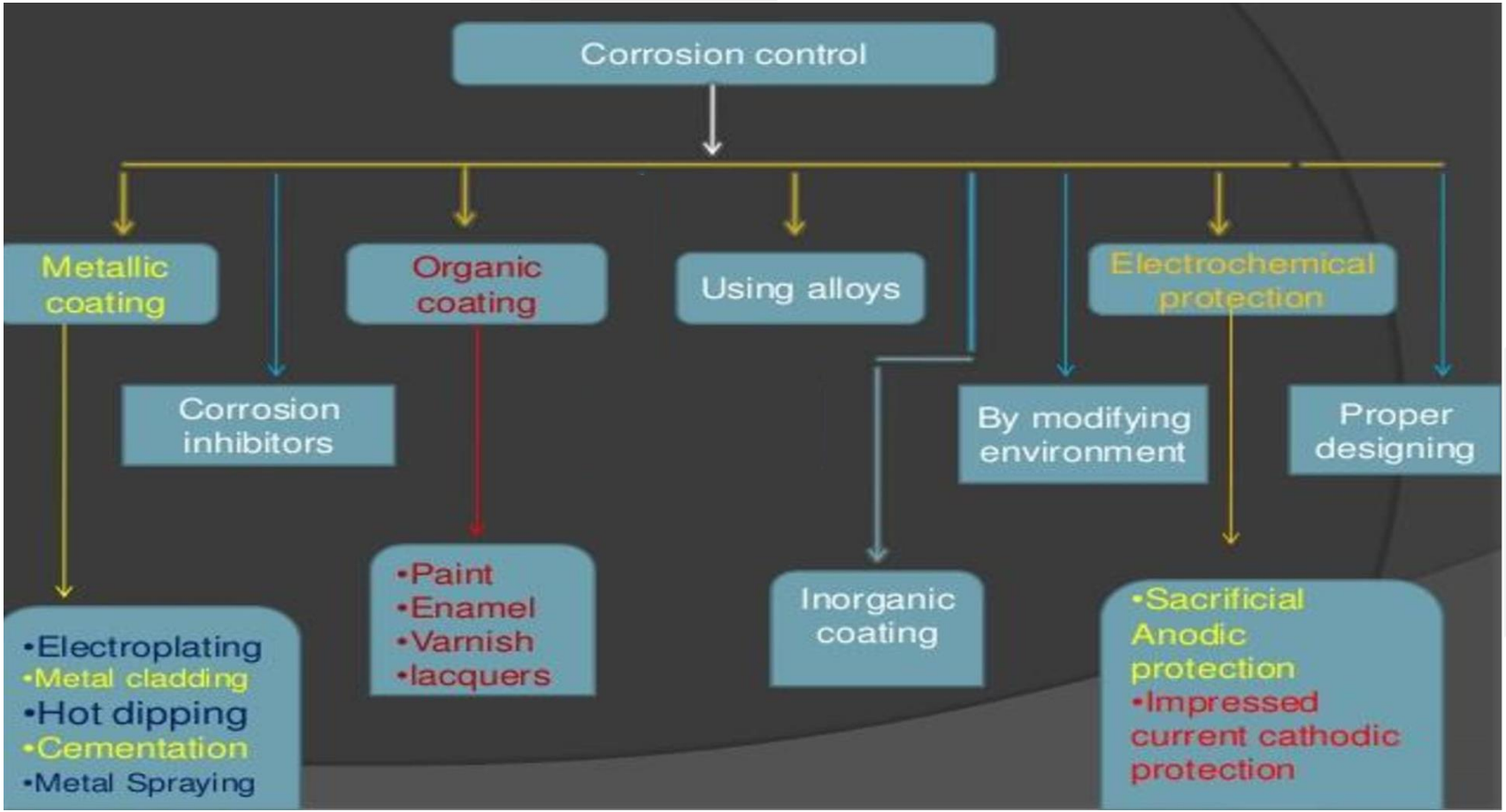
if pH>7, Al_2O_3 film is very reactive, (dissolves) so more corrosion.

(v)Nature of ions present: Chloride ions present in the medium destroy the passive film, while silicate in the medium leads to the formation of insoluble film over the metal surface.

(vi) **Concentration of gas (O_2)**

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Corrosion Control



Metallic Coating

1. Metallic Coating: coating of metal with another metal.

Anodic metallic coating: Galvanization

Cathodic metallic coating: Tinning

Pre-treatment: Cleaning of metal surface

(i) Mechanical cleaning – like Grinding, polishing, sand blasting, rubbing etc.

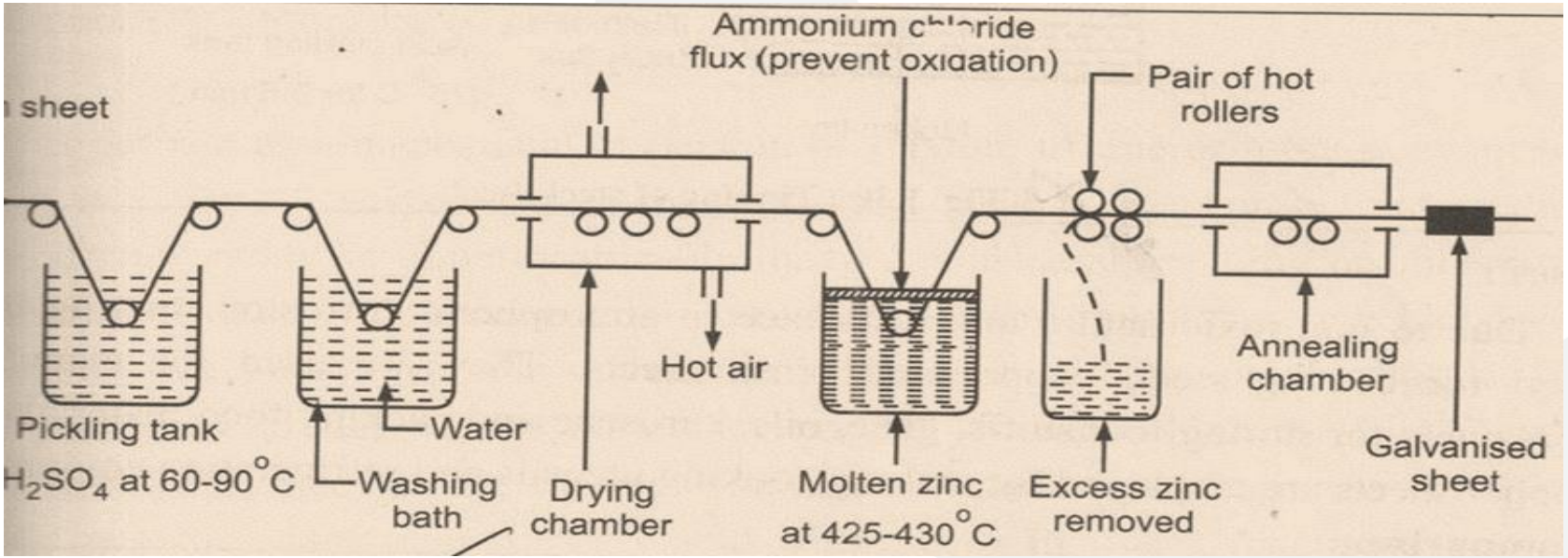
(ii) Removal of oil and grease – As these impurities are organic in nature, they are removed by washing the object with organic solvents like carbon tetrachloride, dichloromethane, chlorobenzene, toluene etc (like dissolves like).

(iii) Removal of Rust- rust is basic in nature and therefore reacts with acid. So rust is removed by dipping the object in hot (around 70 ^{o}C) dilute sulphuric acid (5-10%) solution. This process is also called **PICKLING** and it dissolves rust.

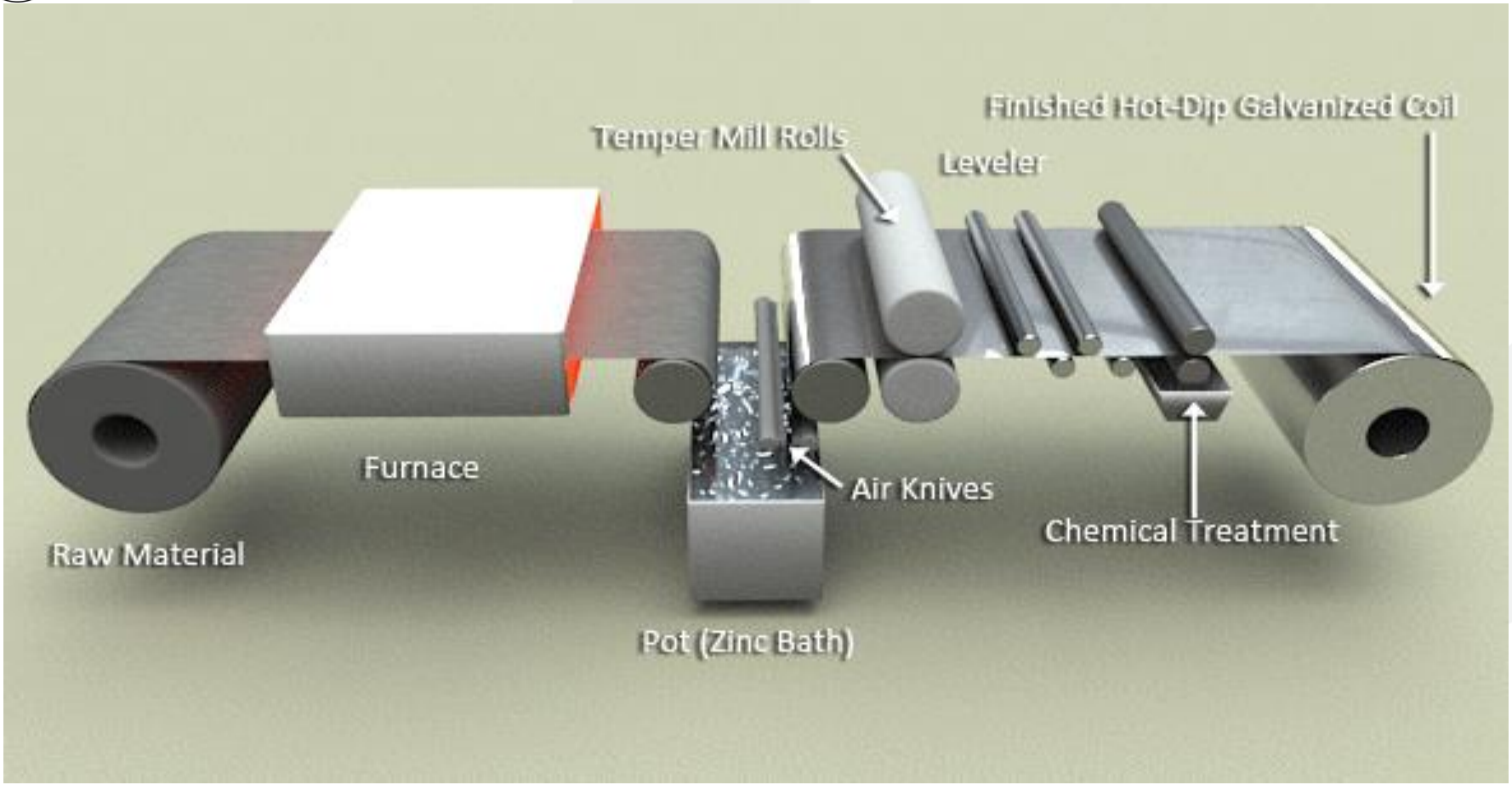
(iv) Washing with water- finally the object is washed with water and then taken for coating.

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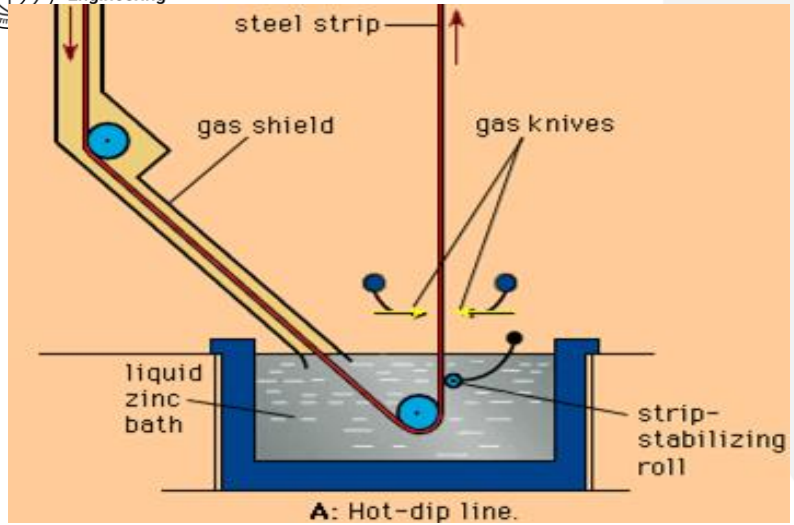
Galvanization (Anodic metallic coating



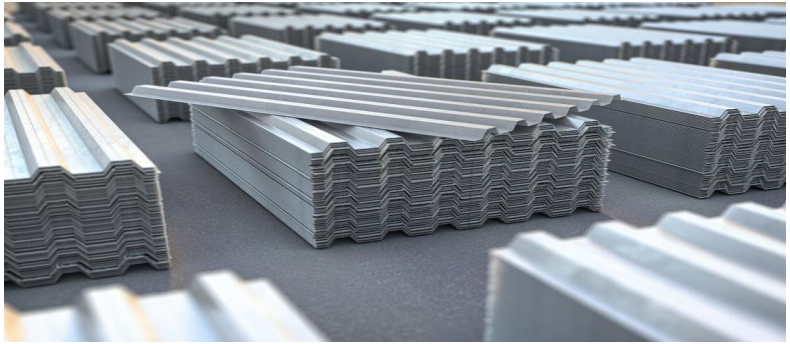
Galvanization process



Galvanized Articles





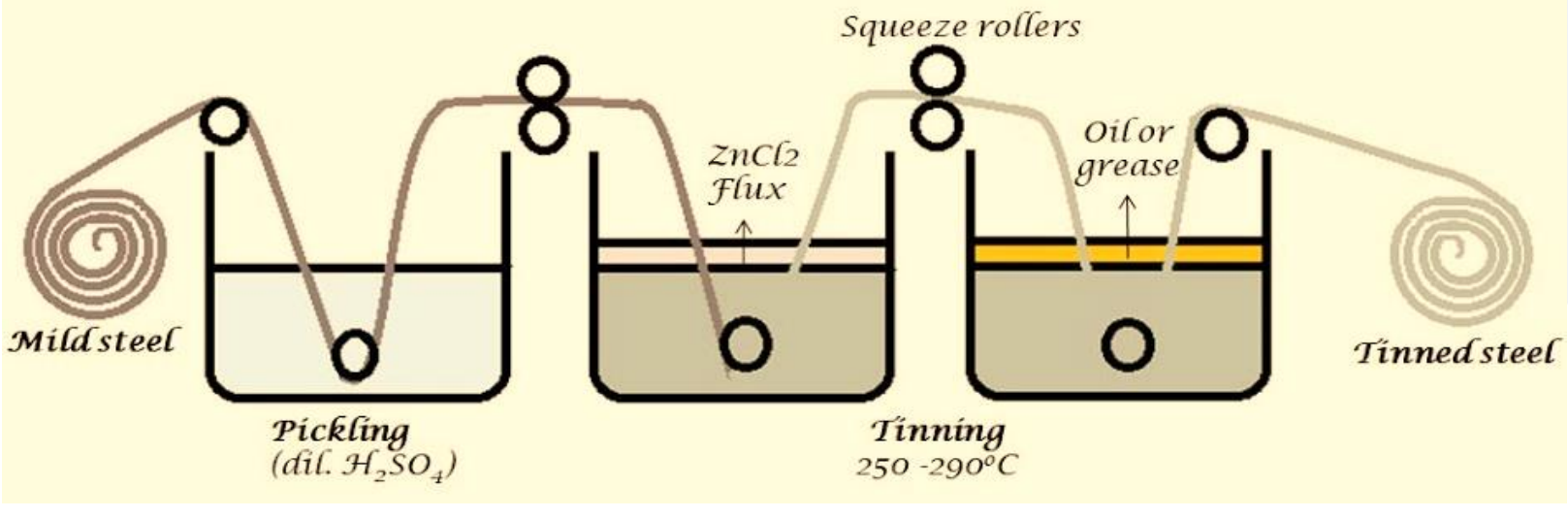






Tinning

- Process of coating tin over iron or steel.
- It is cathodic metallic coating.
- Hot dipping process as tin can be melted very easily at around 220C.
- First object is cleaned, and passed through molten $ZnCl_2$, molten tin and then through oil/grease.



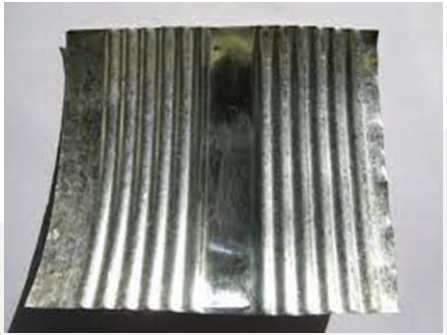
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Tinned Objects

- Tinning is widely used for coating copper and brass sheets used in the manufacture of containers for storing food stuffs.
- Also done on copper vessels used for cooking and on copper wires making them for solderable.









Tinned Objects

- Galvanization is anodic metallic coating and tinning is cathodic metallic coting.
- If there is small crack/hole on the surface of galvanized article, very small base metal (cathode) is exposed to corrosion environment. In such case relative area of anode is larger than cathode, so the rate of corrosion will be minimized.
- Galvanized articles cannot be used for storing food stuffs.

Inorganic Coating

- When the barrier coating is an inorganic compound, then it is considered as inorganic coating.

- Coatings are provided by converting the metal surface into an inorganic compound like oxide or phospahate either by chemical or electrochemical method, hence called as chemical conversion coatings.

- Such coatings in addition to controlling corrosion have the advantages like,

(i) providing electrical insulation property to the surface. (ii)They enhance the adherence of paint and enamel to the surface.

Eg: Phosphating, Aodising.

Phosphating

Phosphating bath consists of,

i. Free phosphoric acid- it is the main constituent. It reacts with the metal and produces metal sulphate.

ii. Metal phosphates of zinc and manganese-they help in increasing the adherence of the phosphate layer to the metal surface. They also decrease the porosity of the coating.

iii. Nitrates, chromates, peroxide etc-they are oxidizing agents. Hence, they help in the oxidation of the metal and thus accelerate the formation of phosphate layer.

$$3Fe + 2H_3PO_4 \longrightarrow Fe_3(PO_4)_2 + 3H_2$$

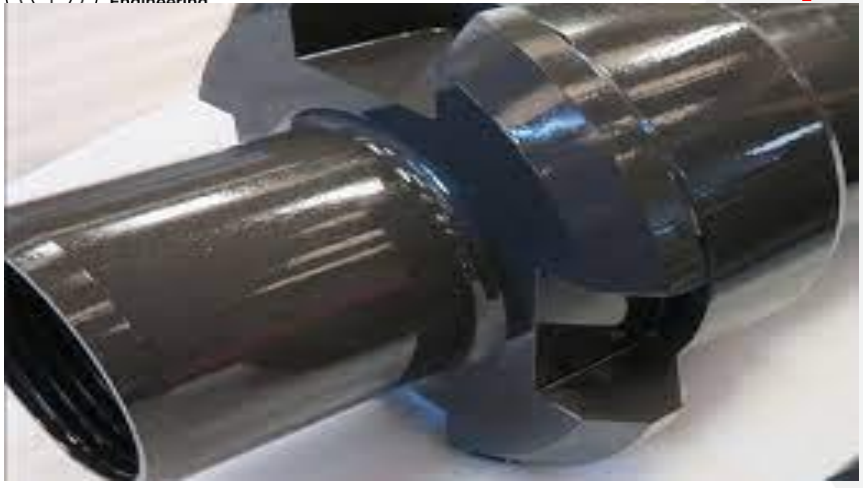
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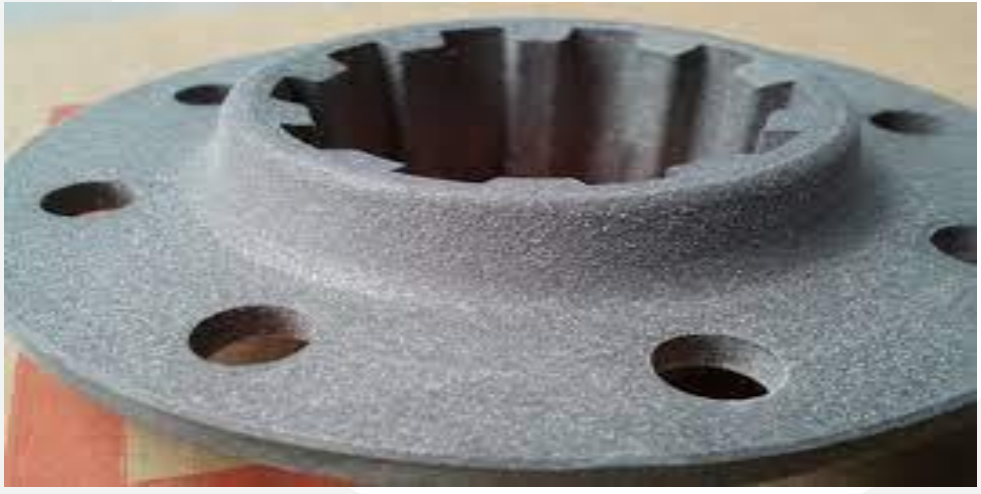
Phosphating



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Phosphated Articles



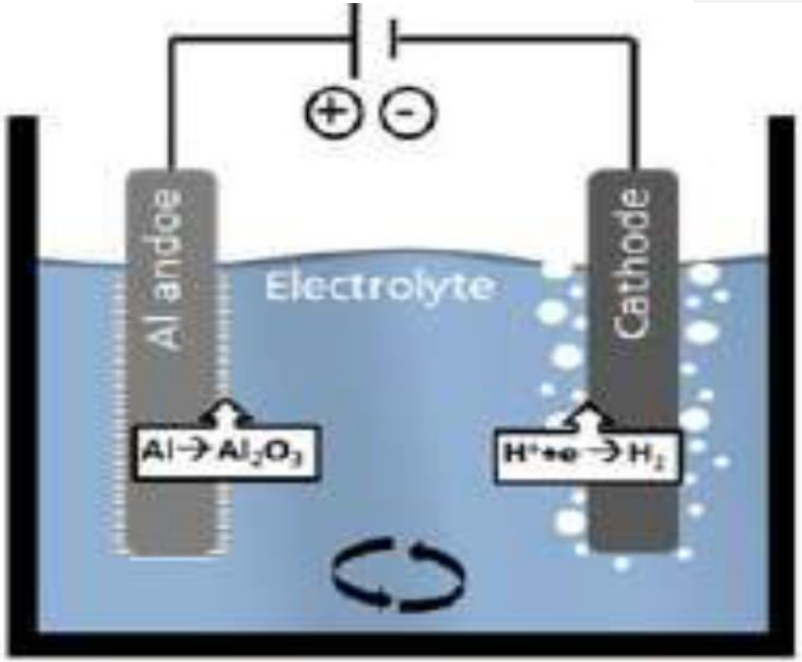


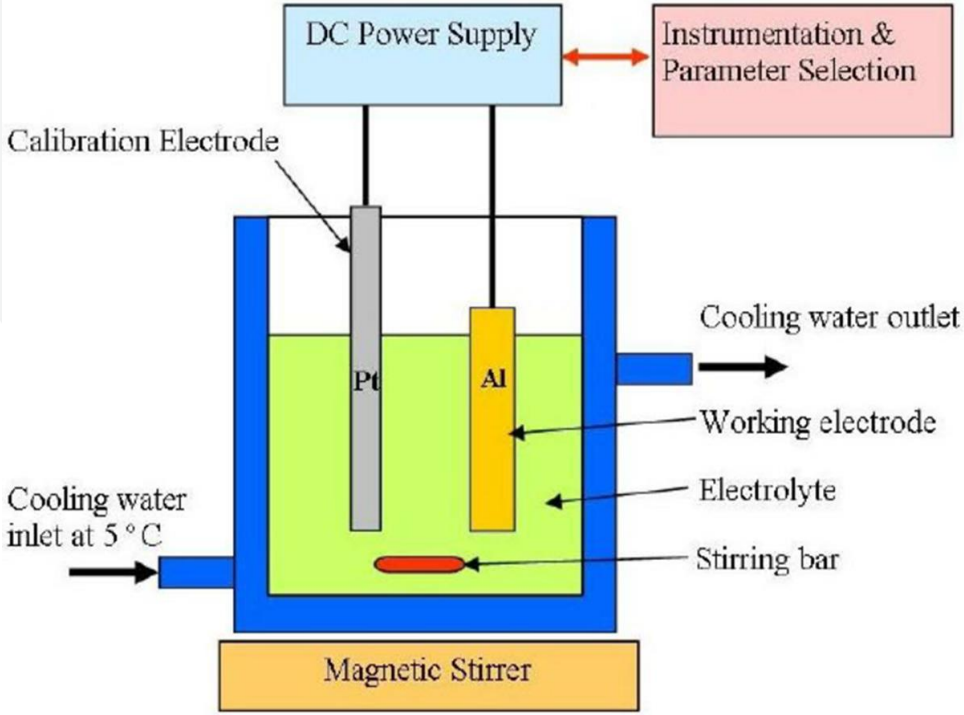




Anodizing of Aluminium

- The method involves developing an oxide layer on the surface by induced oxidation, in an oxidizing environment, making the object anode and passing current.





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Anodizing Process

- **Anodizing –** When the current is flowing in the cell the following sequence of events is believed to occur. Sulphuric acid begins to decompose, the hydrogen ions moving to the cathode where they are reduced to hydrogen gas:

$$2H^{+} + 2e^{-} \rightarrow H_2(g)$$

- Simultaneously, negatively charged anions, i.e. hydroxide, sulphate and maybe oxide ions move to the anode. The electrical charge in the circuit causes positively charged aluminum ions (Al^{3+}) to be generated in the anode and in turn move toward the cathode. At the anode surface they react with the oxide/hydroxide ions to form aluminium oxide.

At anode: $2\ Al + 3\ H_2O \rightarrow\ Al_2O_3 + 6\ H^{+} + 6e^{-}$

At cathode: $6\ H^{+} + 6\ e^{-} \rightarrow\ 3H_2\uparrow$

Over all reaction: $2\ Al + 3H_2O \rightarrow Al_2O_3 + 3\ H_2\uparrow$

Sealing -$Al_2O_3.H_2O$

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Anodized Articles







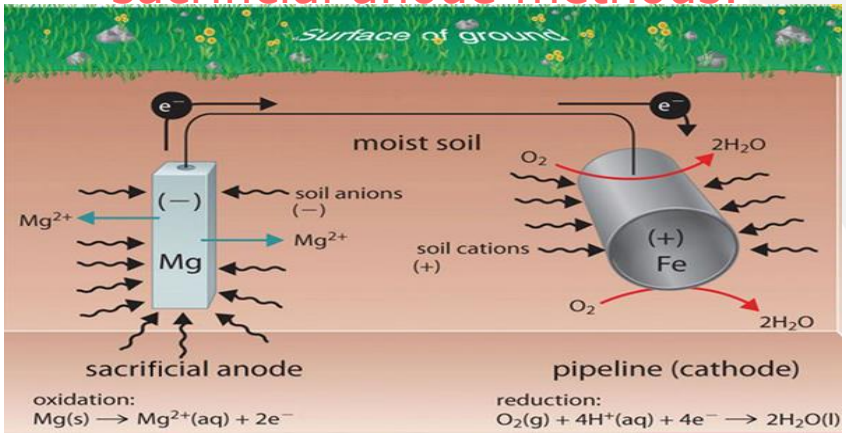
Cathodic Protection

- Cathodic protection (CP) is a technique used to control the corrosion of a metal surface by making it the cathodic side of an electrochemical cell.
- In this method the metal to be protected from corrosion is converted into cathode by connecting into a metal which is anodic to it.
- The base metal is usually iron, copper or brass. Metals like Mg, Al and Zn are used as anodes.
- These metals being more active acts as anode undergo corrosion and supply electrons to the specimen.
- In this way the protected metals act as cathode.
- Since anodic metals are scarified to protect the metal structure this technique is called sacrificial anode method.

Cathodic Protection

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- Example: Mg block connected to buried oil storage tanks.
- Zn wire fixed to the sides of ocean-going ships are the best examples of sacrificial anode methods.









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Corrosion Testing

Weight loss method

- Weight-loss tests are mostly used for measurement of corrosion.
- In this, clean metal coupon or a standard test piece is measured, weighed, and exposed to the corrosion environment for a known time.
- The piece is then taken out, cleaned to remove the corrosion products and reweighed.
- The rate of corrosion of the metal, R is then calculated from the equation.

Corrosion Testing

- The rate of corrosion of the metal, R is then calculated from the equation.

$$CPR = KW / ATD,$$

- Where K is constant,

T is time of exposure to the nearest 100th of an hour.
W is the weight loss to the nearest milligram
A is the area to the nearest sq.cm, and
D is density, g/cm^3.
CPR is corrosion penetration rate

- The result is expressed in many different units such as $mg/dm^2/day$ or $oz/ft^2/day$ or month or millimeters per year (MMPY) or Miles per year (MPY). Assuming that the surface corrosion was uniform, these units may be converted to depth of corrosion or penetration.

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Corrosion Testing

Problem-1

An alloy of thickness 100 $inch^2$ is exposed to corrosion environment for the period of one year. After one year its weight was found to be 35g due to corrosion. Calculate its CPR in mmpy and mpy. Given density of alloy= 8.4 g/cm^3, 1$inch^2$=6.45cm^2, K= 87.36 for mmpy and 534 for mpy.

	Given value	mmpy	mpy
K		87.6	534
W	35g	35*1000mg	35*1000mg
D	8.4 g/cm^3	8.4 g/cm^3	8.4 g/cm^3
A	100 $inch^2$	100*6.45 cm^2	100 $inch^2$
T	1 year	365*24 hrs	365*24 hrs

CPR = KW / ATD

CPR=0.064 mmpy

CPR=2.54 mpy

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Corrosion Testing

Problem-2

A sheet of carbon steel 1m wide and 3m long has lost 40g to corrosion over the past 6 months. Calculate its CPR in mmpy and mpy. Given density of steel= 7.8 g/cm^3, $1inch^2=6.45cm^2$, K= 87.36 for mmpy and 534 for mpy.

$1m^2= 1550\ Inch^2$

	Given value	mmpy	mpy
K		87.6	534
W	40g	40*1000mg	40*1000mg
D	7.8 g/cm^3	7.8 g/cm^3	7.8 g/cm^3
A	1*3 m^2	3*100*100 cm^2	3*1550 $inch^2$
T	6 months	6*30*24 hrs	6*30*24 hrs

CPR = KW / ATD

CPR=0.064 mmpy

CPR=2.54 mpy

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Corrosion Testing

Problem-3

An alloy sheet of area150 $Inch^2$ is exposed to air near ocean. After two years it was found to experience a weight loss of 95g due to corrosion. Calculate its CPR in mmpy and mpy. Given density of alloy= 8.4 g/cm^3, K= 87.36 for mmpy and 534 for mpy.

6.45 cm^2= 1 $Inch^2$

	Given value	mmpy	mpy
K		87.6	534
W	95g	95*1000mg	95*1000mg
D	8.4 g/cm^3	8.4 g/cm^3	8.4 g/cm^3
A	150 $Inch^2$	150*6.45 cm^2	150 $inch^2$
T	2 Years	2*365*24 hrs	2*365*24 hrs

CPR = KW / ATD

CPR=..... mmpy

CPR=..... mpy

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